

QP CODE: 4220100911



Reg No : .....

Name : .....

**M.C.A. DEGREE EXAMINATION, JANUARY 2022**

**First Semester**

**Core - MCACT101 - MATHEMATICAL & STATISTICAL FOUNDATION FOR  
COMPUTER APPLICATIONS**

2020 Admission Onwards

3EE7CF21

Time: 3 Hours

Maximum: 75 Marks

**Part A**

*Answer any **ten** questions*

*Each question carries **3** marks*

1. Define power set. Find the power set of (i)  $\emptyset$  (ii)  $\{\emptyset\}$
2. Explain symmetric and antisymmetric relations with examples.
3. Define a partial ordering with an example
4. Define Biconditional Statement
5. State De Morgan's law in propositional logic
6. Explain universal quantifiers with examples
7. The two regression coefficients are -0.7 and -0.2. Find Correlation coefficient
8. Define equally likely events
9. Can the following be a probability density function. If not justify.  
$$f(x) = \begin{cases} 1/2 & \text{When } x = 0 \\ 2/3 & \text{When } x = 1 \\ 1/4 & \text{When } x = 2 \\ 0 & \text{Otherwise} \end{cases}$$
10. What are the desirable properties of a point estimate?
11. Briefly explain two important applications of t distribution in small sample tests.
12. Explain the procedure of testing of independence of two attributes.





(10×3=30 marks)

### Part B

Answer **all** questions

Each question carries **9** marks

13. a) Prove that the congruence modulo  $m$  is an equivalence relation on the set of integers.

OR

- b) Determine whether the functions  $f$  and  $g$  are bijections from  $\mathbb{R}$  to  $\mathbb{R}$  where  $f(x) = 2x - 3$  and  $g(x) = x^2 + 1$ . Find  $f \circ g$  and  $g \circ f$  ?

14. a) Determine the validity of the following argument H1: If Canada is a country then New York is a city, H2 : New York is a City, C: Canada is a country

OR

- b) Show that  $\exists x, M(x)$  follows logically from  $\forall x (H(x) \rightarrow M(x))$ , and  $\exists x, H(x)$

15. a) The two regression equations are given by  $8x - 10y + 66 = 0$  and  $40x - 18y - 214 = 0$

- Identify the regression lines of  $y$  on  $x$  and  $x$  on  $y$
- obtain regression coefficients and the correlation coefficient
- Find the standard deviation of  $y$  if the standard deviation of  $x$  is 4

OR

- b) The chance that doctor A will diagnose the disease B correctly is 60%. The chance that the patient will die by his treatment after correct diagnosis is 40% and the chance of death by wrong diagnosis is 70%. A patient of Doctor A who had disease B died, What is the probability that his disease was correctly diagnosed

16. a) A car hire firm has two cars which it hires out day by day. The number of demands for a car on each day is distributed as a Poisson variate with mean 1.5. Calculate the proportion of days on which

- neither car is used
- some demand is refused

OR

- b) In a distribution exactly normal 7% of the items are under 35 and 89% are under 63. What are the mean and standard deviation of the distribution?

17. a) 12 rats were given a high protein diet and another set of 7 rats given a low protein diet. The gain in weight in gms observed in the two sets are given below:

High protein diet : 13 14 10 11 12 16 10 8 11 12 9 12

Low protein diet : 7 11 10 8 10 13 9

Examine whether the high protein diet is superior to the low protein diet at 5% level of significance.

OR





- b) An examination was given to 50 students at college A and to 60 students at college B .At A the mean grade was 75 with S.D of 9 and at B the mean grade was 79 with S.D of 7 . Is there significant difference between the performance of the students at A and those at B given that (  $\alpha = 0.05$  )

(5×9=45 marks)

